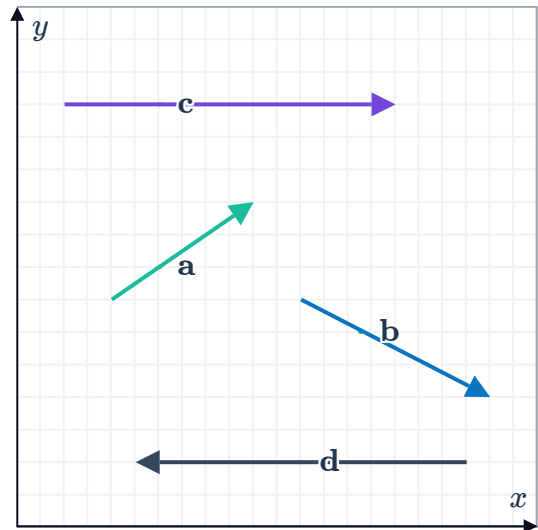
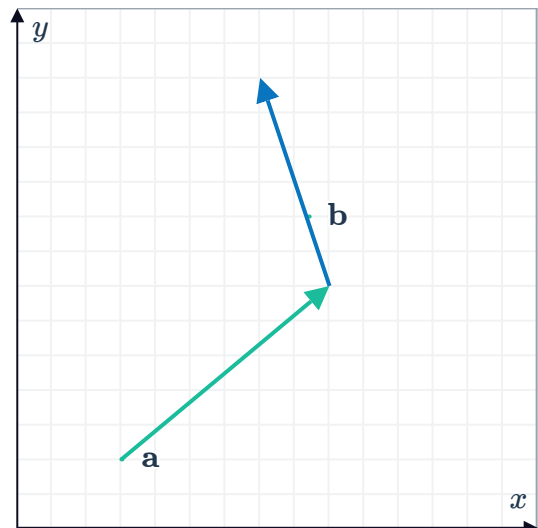


Geometric addition and subtraction of vectors (Extended)

- 1 Consider the following graph and identify which vector is the result of $\mathbf{a} + \mathbf{b}$:



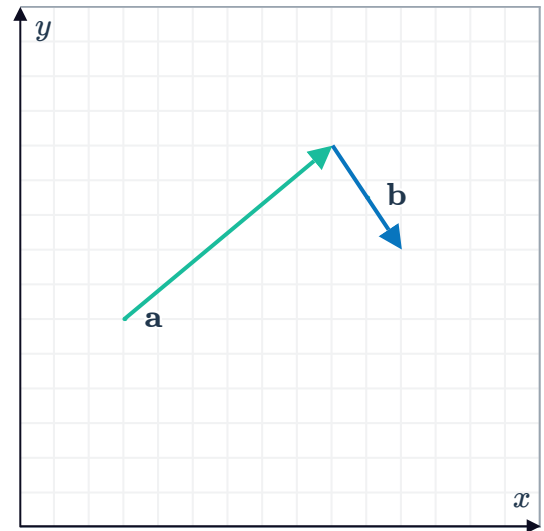
- 2 Consider the graph of vectors $\mathbf{a}$ and $\mathbf{b}$. Plot the result of $\mathbf{a} + \mathbf{b}$.



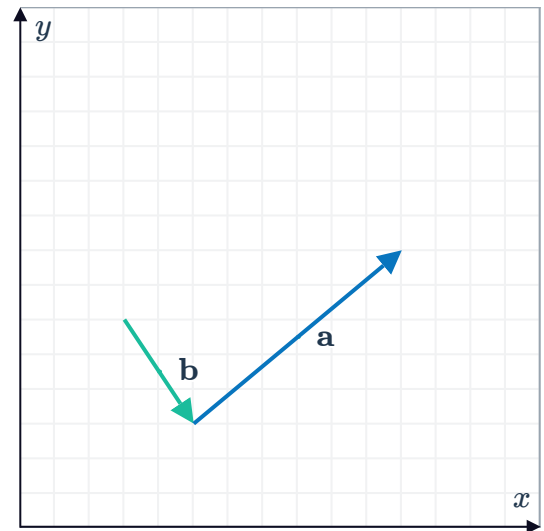
3 Consider the following graphs of **a** and **b**:



a Plot the result of $\mathbf{a} + \mathbf{b}$ in the first graph:



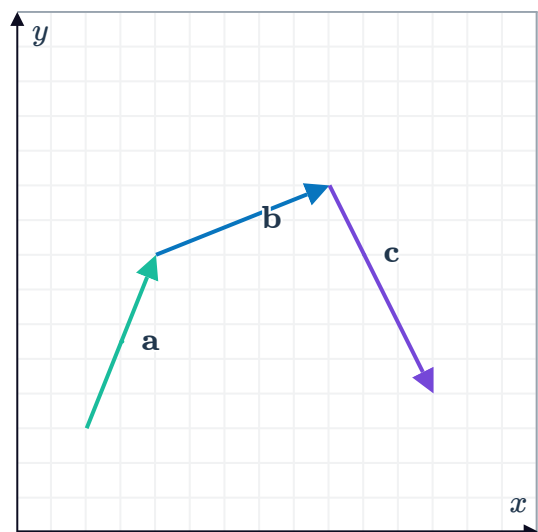
b Plot the result of $\mathbf{b} + \mathbf{a}$ in the second graph:



c Are the two resulting vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{b} + \mathbf{a}$ equal?

4 The vectors **a**, **b** and **c** have been graphed as shown:

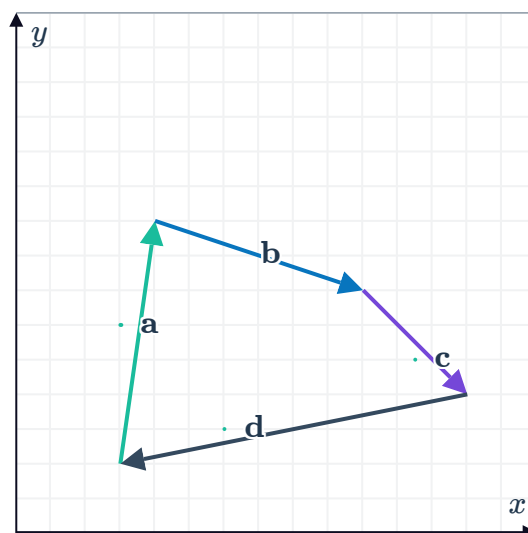
Plot the result of $\mathbf{a} + \mathbf{b} + \mathbf{c}$ on the same axes.



- 5 Vectors **a**, **b**, **c** and **d** have been graphed as shown:



Plot the result of $\mathbf{b} + \mathbf{c}$ on the same axes.



Algebraic addition and subtraction of vectors

- 6 Rewrite $(6, 5) + (7, -4)$ as a single vector.

- 7 For each of the following set of vectors, find:

i $\mathbf{A} + \mathbf{B}$

ii $\mathbf{B} + \mathbf{C}$

iii $\mathbf{A} + \mathbf{B} + \mathbf{C}$

a $\mathbf{A} = \begin{pmatrix} 4 \\ 9 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 8 \\ 2 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 7 \\ 5 \end{pmatrix}$

b $\mathbf{A} = \begin{pmatrix} -6 \\ 7 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} -4 \\ -8 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$

c $\mathbf{A} = \begin{pmatrix} 4.7 \\ -6.2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 7.4 \\ 2.3 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} -5.1 \\ -3.5 \end{pmatrix}.$

d $\mathbf{A} = \begin{pmatrix} 5 \\ 6 \\ 6 \\ 5 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} -2 \\ -5 \\ 7 \\ -5 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 3 \\ 2 \\ 4 \\ -5 \end{pmatrix}.$

Scalar multiplication of vectors

- 8 Let $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, find:



9 Let $\mathbf{a} = \begin{pmatrix} 10 \\ 16 \end{pmatrix}$, find:

a $3\mathbf{a}$

b $-10\mathbf{a}$

c $\frac{1}{2}\mathbf{a}$

d $-\frac{1}{4}\mathbf{a}$

10 Let $\mathbf{a} = \begin{pmatrix} -4 \\ 9 \end{pmatrix}$, find:

a $3\mathbf{a}$

b $-2\mathbf{a}$

c $\frac{1}{2}\mathbf{a}$

d $-10\mathbf{a}$

Scalar multiplication of vectors (Extended)

11 Let $\mathbf{a} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$. Using the origin as the starting point, plot the following vectors:

a $3\mathbf{a}$

b $-2\mathbf{a}$

c $\frac{1}{2}\mathbf{a}$

12 Consider vectors $\mathbf{a}$ and $\mathbf{b}$ plotted on the graph below. Find the column vector for each of the following:

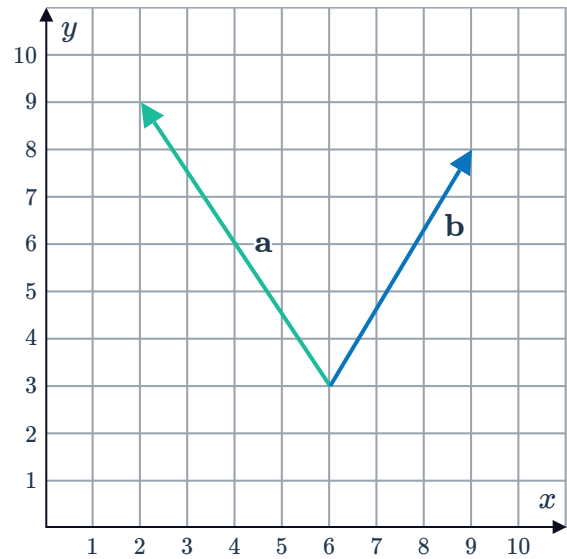
a $\mathbf{a} + \mathbf{b}$

b $3\mathbf{b}$

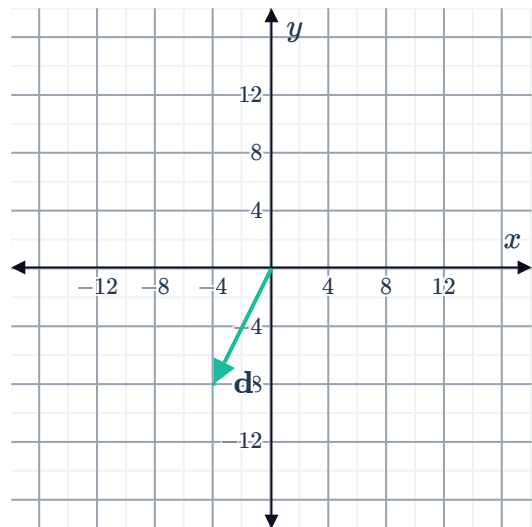
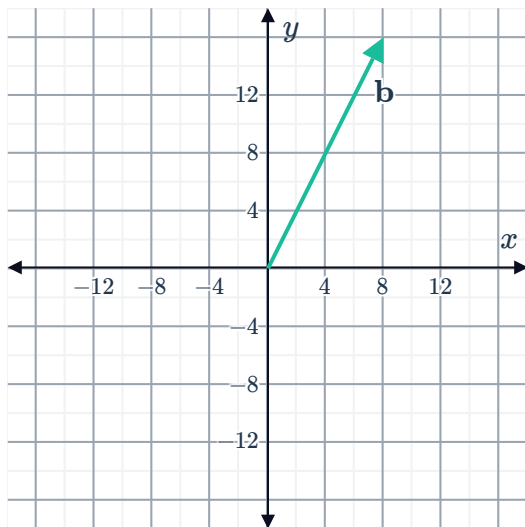
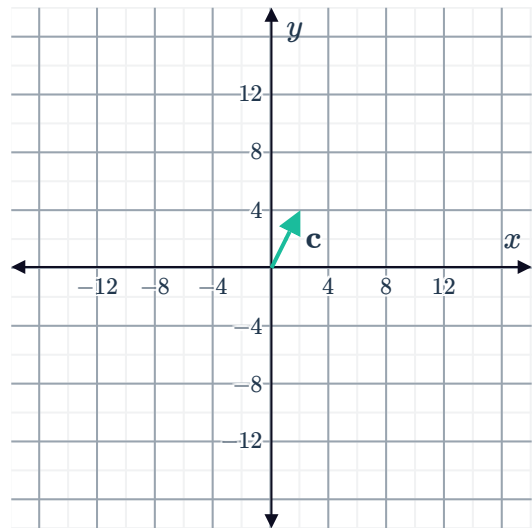
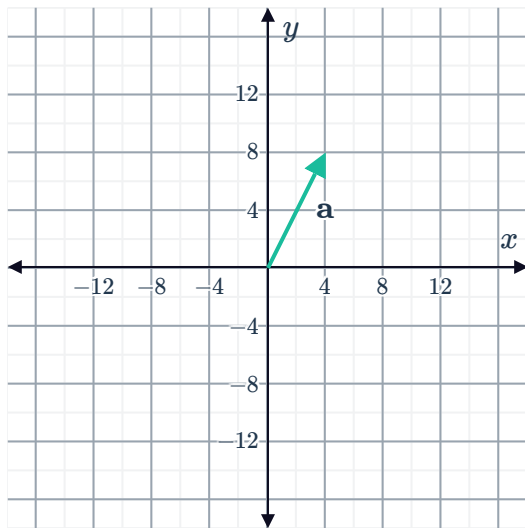
c $4\mathbf{a}$

d $5\mathbf{b} - \mathbf{a}$

e $3\mathbf{a} - 4\mathbf{b}$



13 Consider the vectors shown below:

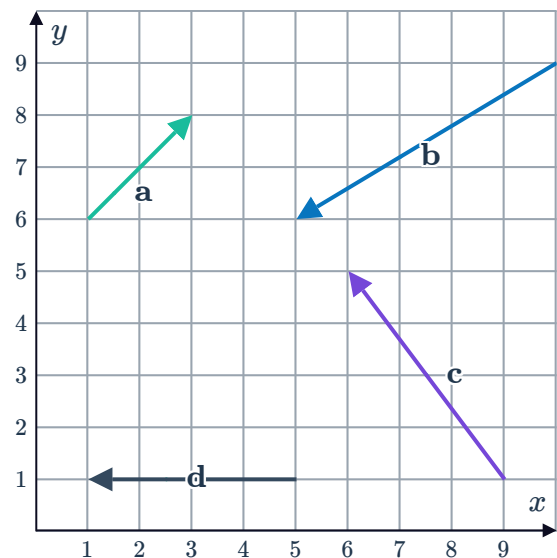


- a Which vector is equivalent to $2\mathbf{a}$?
- c Which vector is equivalent to $-\mathbf{a}$?

- b Which vector is equivalent to $\frac{1}{4}\mathbf{b}$?
- d Which vector is equivalent to $-\frac{1}{2}\mathbf{d}$?

14 Consider the vectors plotted on the graph. Find the column vector for each of the following:

- | | |
|------------------------------|--------------------------------|
| a $3\mathbf{a}$ | b $4\mathbf{c}$ |
| c $2\mathbf{b} + \mathbf{d}$ | d $\mathbf{c} - \mathbf{d}$ |
| e $3\mathbf{a} - \mathbf{b}$ | f $-4\mathbf{b} + 2\mathbf{c}$ |



15 If $\mathbf{a} = (2x, 8x)$, $\mathbf{b} = (3x, 4x)$, and $|\mathbf{a} + \mathbf{b}| = \sqrt{10}x$ find the value of x .

16 If $\mathbf{a} = (2x, 4)$, $\mathbf{b} = (3x, 8x)$, and $|\mathbf{a} + \mathbf{b}| = 13x$, find the value of x .